

Example CNN Architectures

- LeNet-5
 - Convolution and MaxPool layers
- AlexNet
 - SoftMax
 - ReLU activation
 - Dropouts
- ResNet
- Object Detection

AlexNet

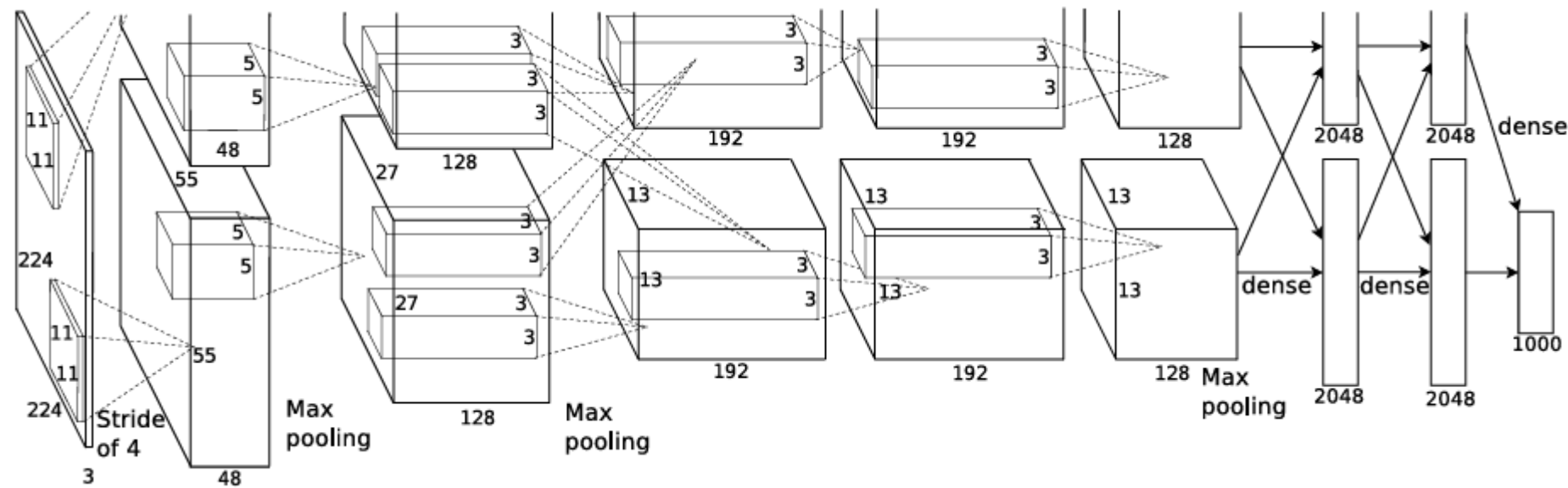
- $\approx 60 \times 10^6$

parameters

- $\approx 700,000$ neurons

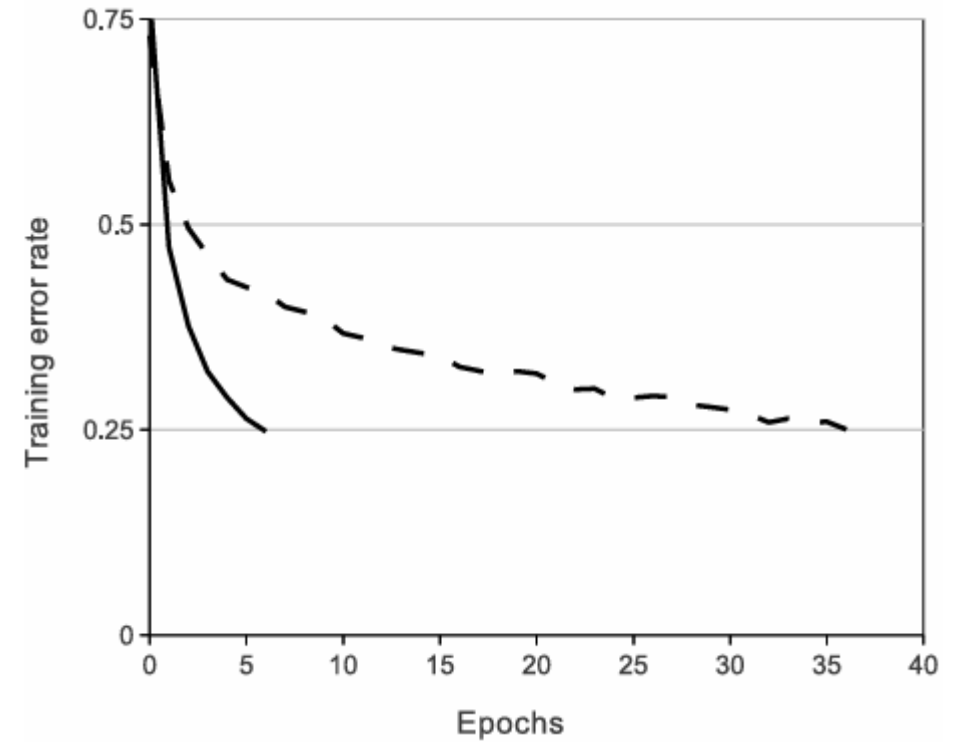
- Trained over 2 GPUs

- Each GPU only has half the weights
- (GTX 580) GPUs can read from each other's memory directly
 - Still not very efficient
 - Communication *across* GPUs is limited.



AlexNet

- ReLU activation
 - $f(x) = \max(0, x)$
 - Reduces training time with GD
 - Faster than $\tanh(\cdot)$ and Logistic
 - “Saturating” vs “Non-saturating” nonlinearities
- Solid line: training error with ReLU
 - $\approx 6 \times$ faster than dotted line: $\tanh(\cdot)$
 - Dataset: CIFAR-10



Reducing Overfitting

- Dropouts

- Randomly remove neurons in the network
- *Independently*
- Including connections to and from

- Model averaging

- “*Dropout: A Simple Way to Prevent Neural Networks from Overfitting,*” JMLR 2014.

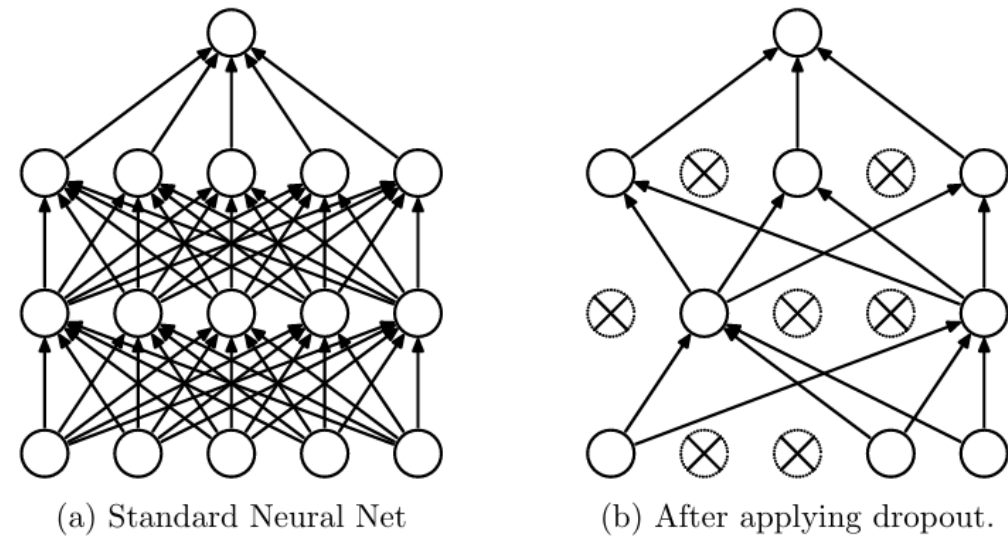


Figure 1: Dropout Neural Net Model. **Left:** A standard neural net with 2 hidden layers. **Right:** An example of a thinned net produced by applying dropout to the network on the left. Crossed units have been dropped.

Reducing Overfitting

- Dropouts

- Randomly remove neurons in the network

- Independently*

- Including connect from

- Model averaging

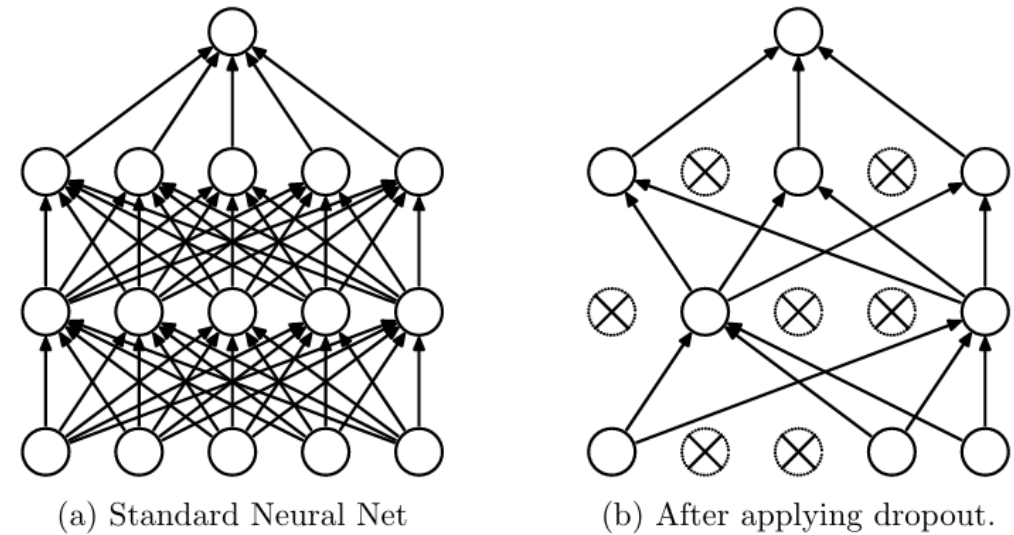


Figure 1: Dropout Neural Net Model. **Left:** A standard neural net with 2 hidden layers. **Right:** An example of a thinned net produced by applying dropout to the network on the left. Crossed units have been dropped.

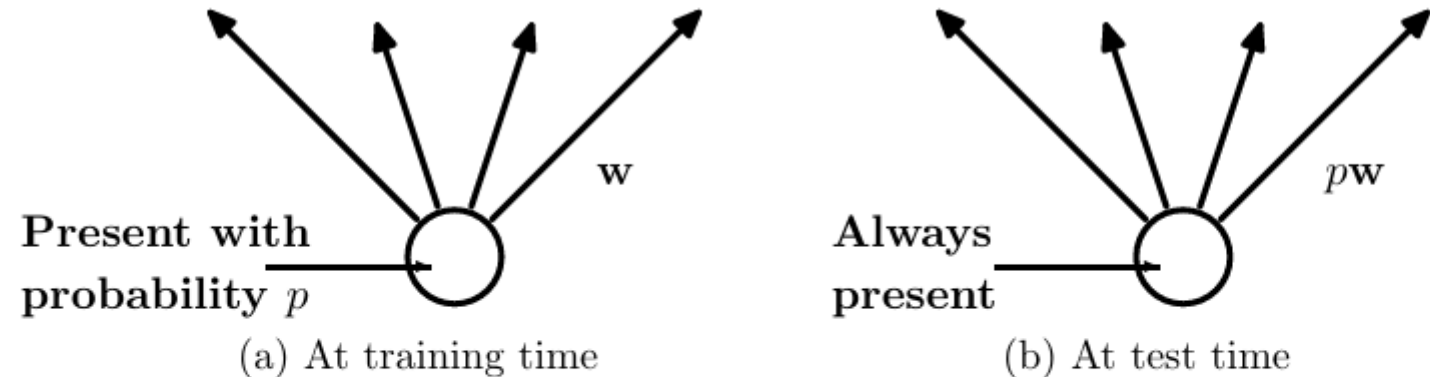


Figure 2: **Left:** A unit at training time that is present with probability p and is connected to units in the next layer with weights $\mathbf{w}$. **Right:** At test time, the unit is always present and the weights are multiplied by p . The output at test time is same as the expected output at training time.

Reducing Overfitting

- Dropouts

- Randomly remove neurons in the network
- *Independently*
- Including connections to and from

- Model averaging

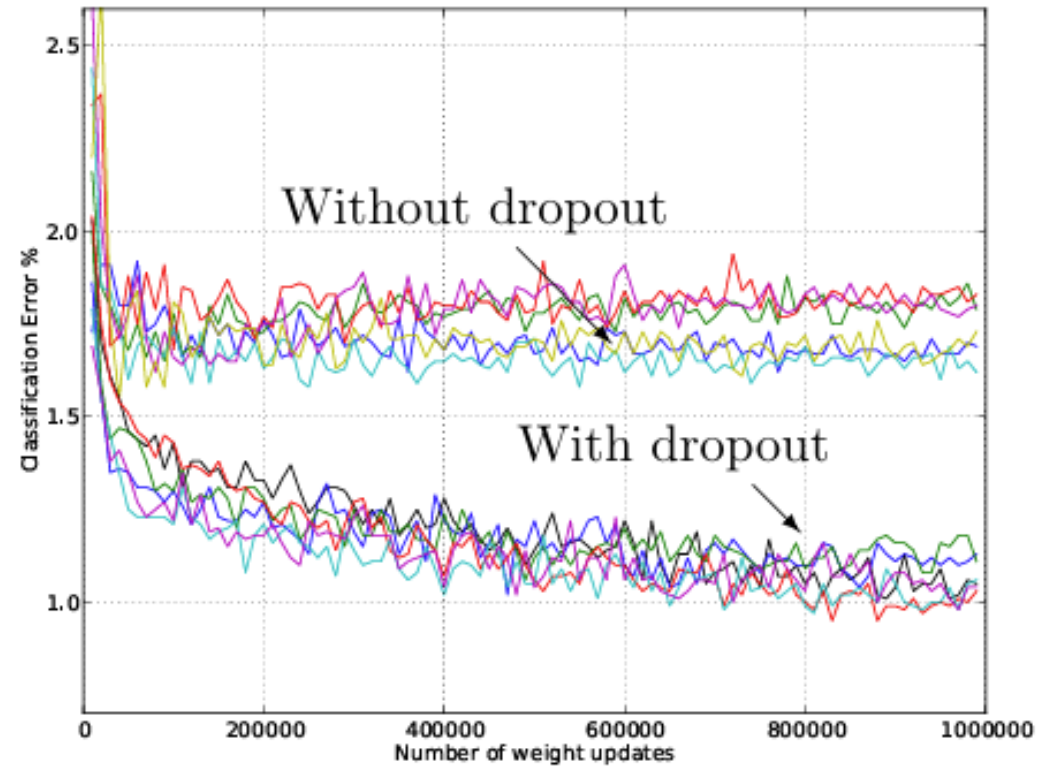
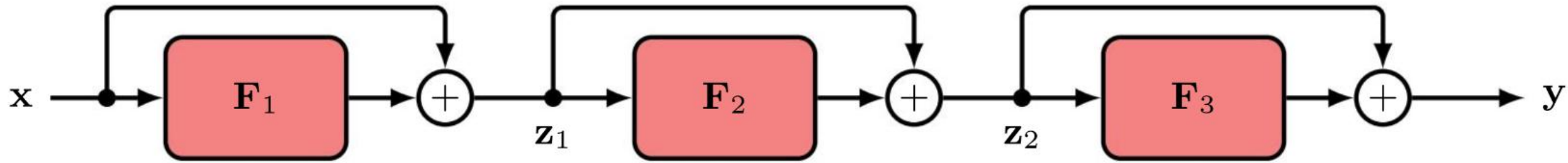


Figure 4: Test error for different architectures with and without dropout. The networks have 2 to 4 hidden layers each with 1024 to 2048 units.

Residual Connections



$$z_1 = F_1(x) + x$$

$$z_2 = F_2(z_1) + z_1$$

$$z_3 = F_3(z_2) + z_2$$

$$\begin{aligned} y &= F_3(z_2) + z_2 \\ &= F_3(F_2(z_1) + z_1) + F_2(z_1) + z_1 \\ &= F_3(F_2(F_1(x) + x) + F_1(x) + x) + F_2(F_1(x) + x) \\ &\quad + F_1(x) + x \end{aligned}$$

Object Detection

- “You Only Look Once,”
 - Note: Current stable version [YOLOV12](#).

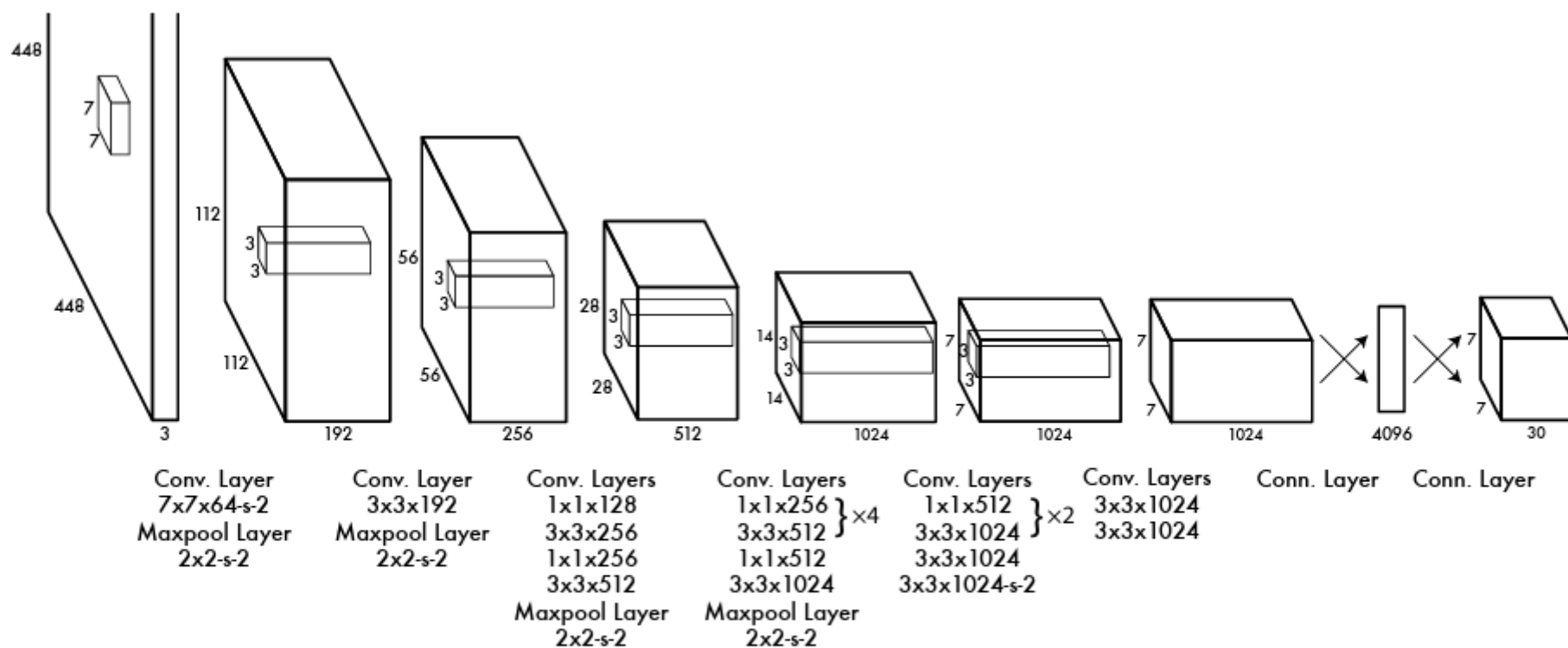


Figure 3: The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.